

Inverse Parametric Sequence Alignment^{*}

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Abstract. We consider the inverse parametric sequence alignment problem, where a sequence alignment is given and the task is to determine parameter values such that the given alignment is optimal at that parameter setting. We describe a $O(mn \log m)$ -time algorithm for inverse global alignment without gaps and a $O(mn \log^2 m)$ time algorithm for global alignment with gaps, where m, n ($n \leq m$) are the lengths of input strings. We then discuss algorithms for local alignment.

1 Introduction

Finding the best alignment of two DNA, RNA or amino acid sequences has become a standard technique for determining similarity between biological sequences. There are hundreds of papers written on this topic and its applications to biology. The review [1] gives relevant references.

Given two sequences S and T of lengths n and m , $n \leq m$, an alignment is obtained by inserting special *space* characters into the two sequences in such a way as to build sequences S' and T' of equal length, denoted by $\mathcal{A} = (S', T')$. A *match* is a position where S' and T' have the same characters. A *mismatch* is a position in which S' and T' have different characters, neither of which is a space. An *indel* is a position in which one of S' and T' has a space. A *gap* is a sequence of one or more consecutive spaces in S' and T' .

An alignment $\mathcal{A}$ can be characterized by its number of matches, mismatches, indels and gaps, denoted w, x, y, z , respectively. In scoring an alignment matches are rewarded, while mismatches, indels and gaps are penalized. Let α, β and γ denote the mismatch, indel and gap penalties. Then the *score* of $\mathcal{A}$ is

$$score_{\mathcal{A}} = w - \alpha x - \beta y - \gamma z$$

The case where the weight of the matches is a parameter is ignored since we can divide all the parameters by this value and reduce it to the above case. The *optimal alignment problem* is to find a maximum-score alignment $\mathcal{A}$ between two strings. For fixed weights, this problem can be solved in $O(nm)$ time [10]. The problem we have just defined is often referred to as *global* alignment, so as to distinguish it from its *local* version, which is defined later (see Section 4).

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The *parametric sequence alignment* problem is to compute optimal alignments for two fixed sequences as a function of varying penalties. The value of an alignment is a linear function of the parameters; thus the parameter space can be partitioned into *optimal regions* such that in every region one alignment is optimal throughout and the regions are maximal for this property. This approach was proposed by Fitch and Smith [4] at first. Later, both mathematical formulations and algorithms for *parametric sequence alignment* were obtained by Gusfield et al. [6, 7]. Additional work is found in [3, 11, 12, 13].

In *inverse parametric optimization* [2] one is given a parametric optimization problem and a desired optimal solution and the task is to determine parameter settings such that the given solution is optimal for those values. The *inverse parametric sequence alignment* problem is to find parameter values such that reference alignment is optimal for those values or, if no such settings exist, find a parameter setting minimizing the numerical difference between the score of the optimal alignment and the score of the reference alignment. These parameter values define an *inverse optimal point* on the parameter space. Inverse parametric computation is useful for deducing parameter settings where the optimal alignment is likely to reconstruct correct alignments that have been determined by other methods.

One way to locate the “correct” parameter settings is to first construct the entire decomposition of the parameter space and then choose the correct values. Alternatively, one can try to find the parameter settings directly. This can be done by gradient descent [7], although it is not clear how to obtain bounds on the worst-case performance of this method. Megiddo’s method of *parametric search* [8, 9] can be used instead, leading to a $O(n^2m^2)$ method for the case where only one parameter is varied. While powerful, Megiddo’s method has the drawback that it leads to complex algorithms. Improvements in the running time are possible by relying on the existence of a *parallel* algorithm for the problem, but this only complicates the results further. Here we give an approach that is much simpler than Megiddo’s method and exploits the integer nature of the scoring of sequence alignments.

Since the optimal regions are bounded by the intersection of hyperplanes, all regions are convex polygons [4, 6, 7, 3]. Hence, the inverse optimal parameter setting(s) must occur at a single vertex (intersection point of three or more optimal regions), at a single edge (intersection line between two optimal regions), or at a single complete polygon of the polygonal decomposition of the parameter space. The main idea of our algorithms is to find the inverse-optimal point in the parameter space using binary search. Our main contribution is a proof that this simple algorithm converges quickly.

The rest of this paper is organized as follows. Section 2 gives a $O(nm \log m)$ algorithm for global alignment without gaps. A $O(nm \log^2 m)$ algorithm for global alignment with gaps is described in Section 3 using a result that is proved in Appendix A. Further results are discussed in Section 4.

2 Global Alignment without Gaps

In this section we consider global alignment where the number of gaps is ignored ($\gamma = 0$). Then, the score function is $score = w - \alpha x - \beta y$. Given a reference alignment $\mathcal{A}_0$ with w_0 matches, x_0 mismatches and y_0 indels, we must find an inverse-optimal point for $\mathcal{A}_0$ in the α, β plane. We need some preliminary results.

Theorem 1 (Gusfield et al. [6]). *Any line forming a boundary between two regions is of the form $\beta = c + (c + 0.5)\alpha$, for some $c > -1/2$.*

Corollary 1. *Suppose (α_0, β_0) is an inverse-optimal point for reference alignment $\mathcal{A}_0$ in the α, β space. Then all points on the line that goes through $(-1, -1/2)$ and (α_0, β_0) are inverse-optimal for $\mathcal{A}_0$.*

Lemma 1 ([6]). *The positive β -axis intersects all the region boundaries.*

Lemma 2 ([6]). *Let $\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_k$ be the optimal alignments encountered by β axis in order of increasing β -value. Then $y_{i+1} < y_i$ for all $\mathcal{A}_i (i < k)$.*

Let $\mathcal{A}_i, \mathcal{A}_j$ be the optimal alignments in two neighboring optimal regions encountered by β axis, with score $w_i - \alpha x_i - \beta y_i$ and $w_j - \alpha x_j - \beta y_j$. Then the equation of the boundary line between the regions is:

$$\beta = \frac{w_i - w_j}{y_i - y_j} + \frac{x_j - x_i}{y_i - y_j} \alpha \quad (1)$$

A *breakpoint* along any given line is the point where the line moves between two adjacent optimal regions.

Lemma 3. *The length of the interval between any two successive breakpoints along the β -axis is greater than $1/m^2$.*

Proof. According to Equation (1), the boundary line of two neighboring optimal regions where $\mathcal{A}_i, \mathcal{A}_j$ are optimal respectively intersects the β axis at $(0, \frac{w_i - w_j}{y_i - y_j})$.

Let $\mathcal{A}_i, \mathcal{A}_j, \mathcal{A}_k$ ($i < j < k$) be the optimal alignments in three consecutive optimal regions when going along the β axis and let $\Delta w_1 = w_j - w_k, \Delta w_2 = w_i - w_j, \Delta y_1 = y_j - y_k, \Delta y_2 = y_i - y_j$. Then the interval between two breakpoints on the β axis is:

$$\Delta\beta = \frac{w_j - w_k}{y_j - y_k} - \frac{w_i - w_j}{y_i - y_j} = \frac{\Delta w_1 \Delta y_2 - \Delta w_2 \Delta y_1}{\Delta y_1 \Delta y_2}$$

Since $\Delta w_1, \Delta w_2, \Delta y_1, \Delta y_2$ are all integers and $\Delta\beta > 0$, then $\Delta w_1 \Delta y_2 - \Delta w_2 \Delta y_1 \geq 1$. Also, $\Delta y_1 + \Delta y_2 = y_i - y_k \leq m + n \leq 2m$, therefore $\Delta y_1 \Delta y_2 \leq ((\Delta y_1 + \Delta y_2)/2)^2 \leq m^2$. It follows that $\Delta\beta > \frac{1}{m^2}$. $\square$

The main idea of our algorithm is to use binary search on the β axis. The details are given below:

Algorithm 1

1. $high = m; low = 0;$
2. **while** $((high - low) > 1/m^2)$
3. $mid = low + (high - low)/2;$
4. compute the optimal global alignment $\mathcal{A}(w, x, y)$ at point $(0, mid);$
5. **if** $(y == y_0)$ **then**
6. **return** the line passing through $(-1, -\frac{1}{2})$ and $(0, mid);$
7. **else**
8. **if** $(y > y_0)$ **then** $low = mid$ **else** $high = mid$ **end if;**
9. **end if**
10. **end while**
11. compute optimal global alignments $\mathcal{A}_{high}, \mathcal{A}_{low}$ for points $(0, high), (0, low);$
12. $mid = low + (high - low)/2;$
13. **if** $(\mathcal{A}_{high}$ is the same as $\mathcal{A}_{low})$ **then**
14. **return** the line passing through $(-1, -\frac{1}{2})$ and $(0, mid);$
15. **else**
16. compute β_0 such that $w_{high} - \beta_0 y_{high} = w_{low} - \beta_0 y_{low};$
17. compute optimal alignment $\mathcal{A}$ for point $(0, \beta_0);$
18. **if** $(w_0 - \beta_0 y_0 == w - \beta_0 y)$ **then**
19. **return** the line passing through $(-1, -\frac{1}{2})$ and $(0, \beta_0);$
20. **else**
21. **return** the line passing through $(-1, -\frac{1}{2})$ and $(0, mid);$
22. **end if**
23. **end if**

Theorem 2. *Algorithm 1 correctly solves the inverse parametric alignment problem for global alignment without gaps in $O(mn \log m)$ time.*

Proof. From equation (1), all breakpoints on the β axis lie below $(0, m)$, so it is correct to restrict the search space to the portion on the β -axis between $(0, 0)$ to $(0, m)$. Lemma 2 and 1 guarantee that binary search works for this problem, since the algorithm can decide to go up or down along the β axis according to the number of indels.

If the algorithm finds a point $(0, \beta_0)$ such that the optimal alignment $\mathcal{A}$ for that point has the same number of indels as the reference alignment, then $(0, \beta_0)$ is either inverse-optimal ($w = w_0$) or it is approximately inverse-optimal ($w \neq w_0$). Following Corollary 1, the algorithm returns a line.

Lemma 3 shows that when the length of the search interval is at most $1/m^2$, it cannot contain a complete optimal region. It includes either part of one optimal region ($\mathcal{A}_{high}$ equals $\mathcal{A}_{low}$) or one breakpoint. In the first case, all points in the remaining search space are approximately inverse-optimal. In the second case, the breakpoint may be inverse-optimal or all points in the remaining search space are approximately inverse-optimal. Thus Algorithm 1 gives the correct answer.

Steps 1, 3, 5-9, 12 need $O(1)$ time; step 11 and 13-22 need $O(mn)$ time; step 4 needs $O(mn)$ time, and the while statement can loop at most $3 \log m$ times. Therefore, the total time is $3 \log m \cdot O(mn) = O(mn \log m)$. $\square$

3 Global Alignment with Gaps

In this section we solve the inverse global alignment problem with gaps. There are now three parameters to consider, α, β and γ . Given a reference alignment $\mathcal{A}_0$ with w_0 matches, x_0 mismatches, y_0 indels and z_0 gaps, we need to find a point on the α, β, γ space where $\mathcal{A}_0$ is optimal or approximately optimal.

First, let us describe the boundary lines of the optimal regions in the α, β, γ space.

Theorem 3 (Gusfield et al. [6]). *Any line forming a boundary between three or more regions is of the form $\beta = c + (c + 1/2)\alpha, \gamma = d + d\alpha$.*

Corollary 2. *Suppose $(\alpha_0, \beta_0, \gamma_0)$ is an inverse-optimal point in the α, β, γ space for reference alignment $\mathcal{A}_0$. Then every point on the line that passes through $(-1, -1/2, 0)$ and $(\alpha_0, \beta_0, \gamma_0)$ is inverse-optimal for $\mathcal{A}_0$.*

Corollary 3. *All region boundaries intersect with either the positive β, γ coordinate plane or with the positive α, γ coordinate plane.*

Finally, as in Equation (1), the boundary line between optimal regions on the β, γ plane associated with alignments $\mathcal{A}_i$ and $\mathcal{A}_j$ has the form

$$\beta = \frac{w_i - w_j}{y_i - y_j} - \frac{z_i - z_j}{y_i - y_j} \gamma \quad (2)$$

A *vertex* is the intersection point of three or more optimal regions. Suppose vertex v is intersection point of optimal regions whose optimal alignments are $\mathcal{A}_1, \mathcal{A}_2, \mathcal{A}_3$. Let $\Delta w_1 = w_1 - w_2, \Delta w_2 = w_2 - w_3$, etc. Then $v = (\beta_v, \gamma_v)$, where

$$\beta_v = \frac{\Delta w_1 \Delta z_2 - \Delta w_2 \Delta z_1}{\Delta y_1 \Delta z_2 - \Delta y_2 \Delta z_1} \quad \text{and} \quad \gamma_v = \frac{\Delta w_1 \Delta y_2 - \Delta w_2 \Delta y_1}{\Delta z_1 \Delta y_2 - \Delta z_2 \Delta y_1} \quad (3)$$

According to Corollary 3, if there is an inverse-optimal point on the search space, then there must be an inverse-optimal point on either the positive β, γ coordinate plane or the positive α, γ coordinate plane. Thus we can search on the β, γ coordinate plane and the α, γ coordinate plane respectively to find an inverse-optimal point or approximately inverse-optimal point. Let us consider search on β, γ coordinate plane first ($\alpha = 0$).

The algorithm uses the following idea to reduce the search space. Let $v = (0, \beta_v, \gamma_v)$ be a point in the remaining search space on the β, γ plane and let $\mathcal{A}_v$ be the optimal alignment at v , with w_v matches, x_v mismatches, y_v indels and z_v gaps. If reference alignment $\mathcal{A}_0$ is optimal at v , then v is an inverse-optimal point. Otherwise, suppose $\mathcal{A}_0$ is optimal at point $(0, \beta, \gamma)$. By the optimality of $\mathcal{A}_0$ and $\mathcal{A}_v$, it follows that:

$$w_0 - \beta y_0 - \gamma z_0 \geq w_v - \beta y_v - \gamma z_v \quad \text{and} \quad w_v - \beta_v y_v - \gamma_v z_v \geq w_0 - \beta y_0 - \gamma z_0.$$

Therefore,

$$(y_v - y_0)\beta + (z_v - z_0)\gamma \geq (y_v - y_0)\beta_v + (z_v - z_0)\gamma_v \quad (4)$$

If $\mathcal{A}_0$ is not an optimal alignment at any point on the β, γ plane, then suppose $(0, \beta, \gamma)$ is an approximately inverse-optimal point that minimizes the numerical difference between the score of the optimal alignment and the score of $\mathcal{A}_0$, and that $\hat{\mathcal{A}}$ is optimal at that point. Thus, we have: $(w_v - \beta_v y_v - \gamma_v z_v) - (w_0 - \beta_0 y_0 - \gamma_0 z_0) \geq (\hat{w} - \beta \hat{y} - \gamma \hat{z}) - (w_0 - \beta y_0 - \gamma z_0)$ and $(\hat{w} - \beta \hat{y} - \gamma \hat{z}) \geq (w_v - \beta y_v - \gamma z_v)$. Therefore $(y_v - y_0)\beta + (z_v - z_0)\gamma \geq (y_v - y_0)\beta_v + (z_v - z_0)\gamma_v$, which is the same as inequality (4).

The boundary line l of the halfplane defined by (4) passes through v , so l divides the remaining search space into two regions. The region whose points do not satisfy inequality (4) can be discarded, thereby reducing the search space. Line l becomes a new boundary line for the remaining search space; we say that this boundary line is *defined by* point v and alignment $\mathcal{A}_v$. From a property of the centroids of convex polygons that is proved in Appendix A (Theorem 7), if the *centroid* of the present search space is selected as point v , the search space will reduce by a factor of at least $4/9$.

Using the centroid as described above, we repeatedly reduce the search region on the β, γ plane until its area is smaller than $1/2m^{10}$. By Theorem 4, which is proved later, when the region is reduced to this size, it cannot include a complete optimal region.

After the binary search terminates, if there exists an inverse-optimal point, there must exist an inverse-optimal *vertex* in the remaining search space. From Lemma 6, proved later, the inverse-optimal vertex cannot occur on the boundary line of the remaining search space. From Theorem 5, also proved below, if there exists inverse-optimal vertex (β^*, γ^*) in the remaining search space, there exist two distinct boundary lines, defined by (β_1, γ_1) and $\mathcal{A}_1$ and by (β_2, γ_2) and $\mathcal{A}_2$, such that $\mathcal{A}_1, \mathcal{A}_2$ that are optimal at (β^*, γ^*) ; that is, (β^*, γ^*) is the intersection of the scores of $\mathcal{A}_0, \mathcal{A}_1$, and $\mathcal{A}_2$. Thus, to locate the inverse optimal point, we consider different pairs of boundary lines defined by alignments $\mathcal{A}_1, \mathcal{A}_2$. For each pair, we compute the point $(\hat{\beta}, \hat{\gamma})$ where $\mathcal{A}_0, \mathcal{A}_1, \mathcal{A}_2$ have equal score and check if it is an inverse-optimal point. If so, we stop; otherwise, we continue.

We now describe the algorithm:

Algorithm 2

1. set the search space $R = \{(0, \beta, \gamma) | 0 \leq \beta \leq m^2, 0 \leq \gamma \leq m^2\}$;
2. **while** ($Area(R) > 1/2m^{10}$)
3. compute the centroid $v(\beta_v, \gamma_v)$ of search space R ;
4. compute the optimal global alignment $\mathcal{A}_v$ at point v on β, γ space;
5. **if** ($\mathcal{A}_v$ is the same as $\mathcal{A}_0$) **then**
6. **return** the line passing through $(-1, -\frac{1}{2}, 0)$ and $(0, \beta_v, \gamma_v)$;
7. **else**
8. $R \leftarrow R \cap$ the halfplane defined by Equation (4);
9. **end if**
10. **end while**
11. store all pairs of different boundary lines of R into stack S ;

12. **while** (S is not empty)
13. pop a pair of boundary lines defined by alignments $\mathcal{A}_1, \mathcal{A}_2$ from S ;
14. compute $(\hat{\beta}, \hat{\gamma})$ so $\mathcal{A}_0, \mathcal{A}_1, \mathcal{A}_2$ have same score at $(\hat{\beta}, \hat{\gamma})$;
15. **if** $((\hat{\beta}, \hat{\gamma})$ is in R) **then**
16. compute the optimal global alignment $\hat{A}$ at point $(\hat{\beta}, \hat{\gamma})$;
17. **if** $(\hat{A}$ and $\mathcal{A}_0$ have same score at $(\hat{\beta}, \hat{\gamma}))$ **then**
18. **return** the line passing through $(-1, -\frac{1}{2}, 0)$ and $(0, \hat{\beta}, \hat{\gamma})$
19. **end if**
20. **end if**
21. **end while**
22. continue to search on α, γ space.

To show that Algorithm 2 is correct, we first need to prove some results.

Lemma 4. *The distance between any two vertices on the β, γ plane is greater than $1/m^4$.*

Proof. According to Equation (3), select two different vertices $v(\beta_v, \gamma_v), u(\beta_u, \gamma_u)$,

$$\text{where } \beta_v = \frac{\Delta w_1 \Delta z_2 - \Delta w_2 \Delta z_1}{\Delta y_1 \Delta z_2 - \Delta y_2 \Delta z_1}, \gamma_v = \frac{\Delta w_1 \Delta y_2 - \Delta w_2 \Delta y_1}{\Delta z_1 \Delta y_2 - \Delta z_2 \Delta y_1},$$

$$\beta_u = \frac{\Delta w_3 \Delta z_4 - \Delta w_4 \Delta z_3}{\Delta y_3 \Delta z_4 - \Delta y_4 \Delta z_3}, \gamma_u = \frac{\Delta w_3 \Delta y_4 - \Delta w_4 \Delta y_3}{\Delta z_3 \Delta y_4 - \Delta z_4 \Delta y_3}, \text{ then}$$

$$|uv| = \sqrt{(\beta_v - \beta_u)^2 + (\gamma_v - \gamma_u)^2}$$

Since $\Delta w_i, \Delta y_i, \Delta z_i$ are integers that lie between $-m$ and m , when $|uv| > 0$, $|uv| > 1/m^4$. $\square$

Lemma 5. *Suppose θ is the intersection angle between two boundary lines of optimal regions on the β, γ plane, then if $\theta \neq \pi$, $\sin \theta > 1/m^2$.*

Proof. Suppose two boundary lines of optimal regions intersect each other. According to Equation (2) the slopes of the two lines are $k_1 = -\Delta z_1/\Delta y_1, k_2 = -\Delta z_2/\Delta y_2$ respectively. Let θ be the intersection angle between these two boundary lines. Since $\tan \theta = (k_1 - k_2)/(1 + k_1 k_2)$,

$$\sin \theta = \sqrt{\frac{\tan^2 \theta}{1 + \tan^2 \theta}} = \frac{|\Delta y_2 \Delta z_1 - \Delta y_1 \Delta z_2|}{\sqrt{(\Delta y_2 \Delta z_1 - \Delta y_1 \Delta z_2)^2 + (\Delta y_1 \Delta y_2 - \Delta z_1 \Delta z_2)^2}}$$

Since $\Delta y_i, \Delta z_i$ are integers between $-m$ and m , when $\sin \theta > 0$, we have $\sin \theta > \frac{1}{m^2}$. $\square$

Theorem 4. *The area of any complete optimal region on the β, γ plane is greater than $1/2m^{10}$*

Proof. A complete optimal region is composed of at least three vertices. Suppose the minimal complete optimal region is a triangle. By Lemmas 4 and 5, each edge of this triangle has length greater than $\frac{1}{m^4}$ and the sine of each angle is greater than $\frac{1}{m^2}$. Therefore, the area is greater than $\frac{1}{2} \cdot \frac{1}{m^4} \cdot \frac{1}{m^4} \cdot \frac{1}{m^2} = \frac{1}{2m^{10}}$. $\square$

The precondition for the following results is that the area of the remaining search space is smaller than $1/2m^{10}$.

Lemma 6. *The inverse-optimal vertex cannot be on the boundary of the remaining search space.*

Proof. Assume that the reference alignment is $\mathcal{A}_0(w_0, y_0, z_0)$, that the inverse-optimal vertex (β^*, γ^*) is on a boundary line l of the remaining search space, and that the boundary line l is defined by (β_v, γ_v) and alignment $\mathcal{A}_v(w_v, y_v, z_v)$. Notice that $\mathcal{A}_0$ is not optimal at (β_v, γ_v) , then $(\beta^*, \gamma^*) \neq (\beta_v, \gamma_v)$. Since (β^*, γ^*) is on l , according to Equation (4), we have:

$$(y_v - y_0)\beta^* + (z_v - z_0)\gamma^* = (y_v - y_0)\beta_v + (z_v - z_0)\gamma_v \quad (*)$$

By the optimality of $\mathcal{A}_0$ and $\mathcal{A}_v$, we have:

$$w_0 - \beta^*y_0 - \gamma^*z_0 \geq w_v - \beta^*y_v - \gamma^*z_v \quad \text{and} \quad w_v - \beta_vy_v - \gamma_vz_v > w_0 - \beta_vy_0 - \gamma_vz_0.$$

Therefore,

$$(y_v - y_0)\beta^* + (z_v - z_0)\gamma^* > (y_v - y_0)\beta_v + (z_v - z_0)\gamma_v \quad (**)$$

Equations (*), (**) are contradictory, so the assumption is wrong. Thus, the inverse-optimal vertex cannot be on any boundary of the remaining search space. $\square$

Lemma 7. *Suppose reference alignment $\mathcal{A}_0(w_0, y_0, z_0)$ is optimal at vertex (β^*, γ^*) in the remaining search space. Let l be a boundary line of the remaining search space defined by (β_v, γ_v) and alignment $\mathcal{A}_v$. If $\mathcal{A}_v$ is not optimal at (β^*, γ^*) , then the distance between (β^*, γ^*) and l is greater than $\frac{1}{m^8}$.*

Proof. According to inequality (4), the boundary line l is

$$(y_v - y_0)\beta + (z_v - z_0)\gamma = (y_v - y_0)\beta_v + (z_v - z_0)\gamma_v \quad (*)$$

Suppose $\mathcal{A}_v$ is not optimal at (β^*, γ^*) . From the optimality of $\mathcal{A}_0$ and $\mathcal{A}_v$, we have:

$$w_0 - \beta^*y_0 - \gamma^*z_0 = w_v - \beta^*y_v - \gamma^*z_v + \Delta c_1, \quad \Delta c_1 > 0 \quad (**)$$

$$w_v - \beta_vy_v - \gamma_vz_v = w_0 - \beta_vy_0 - \gamma_vz_0 + \Delta c_2, \quad \Delta c_2 > 0 \quad (***)$$

Adding (**) and (***) we obtain

$$(y_v - y_0)\beta^* + (z_v - z_0)\gamma^* = (y_v - y_0)\beta_v + (z_v - z_0)\gamma_v + \Delta c_1 + \Delta c_2 \quad (****)$$

Equations (*) and (****) define two parallel lines, and (****) passes through (β^*, γ^*) . Thus the distance between (β^*, γ^*) and line l equals the distance between line (****) and line l . Hence the distance is:

$$d = \left| \frac{\Delta c_1 + \Delta c_2}{y_v - y_0} \cdot \frac{y_v - y_0}{z_0 - z_v} \right| = \left| \frac{\Delta c_1 + \Delta c_2}{z_0 - z_v} \right|$$

According to (**), $\Delta c_1 = w_0 - w_v - (y_0 - y_v)\beta^* - (z_0 - z_v)\gamma^* > 0$

Since (β^*, γ^*) is a vertex, according to Equation (3) it is given by an expression of the form: $\beta^* = \frac{\Delta w_1 \Delta z_2 - \Delta w_2 \Delta z_1}{\Delta y_1 \Delta z_2 - \Delta y_2 \Delta z_1}$ and $\gamma^* = \frac{\Delta w_1 \Delta y_2 - \Delta w_2 \Delta y_1}{\Delta z_1 \Delta y_2 - \Delta z_2 \Delta y_1}$.

It is clear that $\Delta c_1 > 1/m^2$, then $d > \left| \frac{1}{m^2} \frac{1}{z_0 - z_v} \right| > \frac{1}{m^3}$. $\square$

Theorem 5. *If there exists an inverse-optimal vertex (β^*, γ^*) in the remaining search space, then there exist two boundary lines that are defined by (β_1, γ_1) , where alignment $\mathcal{A}_1$ is optimal, and (β_2, γ_2) , where alignment $\mathcal{A}_2$ is optimal, and alignments $\mathcal{A}_1, \mathcal{A}_2$ are optimal at (β^*, γ^*) .*

Proof. Suppose reference alignment $\mathcal{A}_0$ is optimal at vertex (β^*, γ^*) in the remaining search space.

Assume that no boundary line that is defined by (β_i, γ_i) , alignment $\mathcal{A}_i$ and $\mathcal{A}_i$ is optimal at (β^*, γ^*) . Then from Lemma 7, the distance between (β^*, γ^*) and any point on the boundary of the remaining search space is greater than $\frac{1}{m^3}$. Hence the area of the remaining search space is greater than $\frac{1}{m^3}$. But the area of the remaining search space is smaller than $\frac{1}{2m^{10}}$, a contradiction. So there exists at least one boundary line that is defined by (β_1, γ_1) , alignment $\mathcal{A}_1$ and $\mathcal{A}_1$ is optimal at (β^*, γ^*) .

If there is only one boundary line that satisfies the above requirements, we can find a contradiction from similar reasoning. Thus there exist at least two boundary lines that satisfy above requirements. $\square$

Now we prove that Algorithm 2 works well for this problem.

Theorem 6. *If there exists an inverse-optimal point on the β, γ coordinate plane, then Algorithm 2 can find it in $O(mn \log^2 m)$ time.*

Proof. According to Equation 3, the maximum coordinate for a vertex is (m^2, m^2) . Thus, we can restrict the search space to $0 \leq \beta \leq m^2, 0 \leq \gamma \leq m^2$. If the algorithm finds an inverse-optimal point (β_v, γ_v) during the binary search, it returns a line. Theorem 4 shows when the area of the remaining search space is smaller than $1/2m^{10}$, it cannot include a complete optimal region. So binary search terminates and the algorithm begins to check the vertices in the remaining region. From Lemma 6 the inverse-optimal vertex cannot occur on the boundary. If there exists an inverse-optimal vertex in the remaining search space, from Theorem 5, step 11-21 can find it. So if there exists inverse-optimal point on β, γ coordinate plane, Algorithm 2 can find that point.

The initial area of search space is m^4 ; the binary search terminates when the area is less than $1/2m^{10}$. Every iteration reduces the area of search space by at least $4/9$ and increases the number of boundary lines by at most 1, so there are $O(\log m)$ iterations and $O(\log m)$ boundary lines. In every iteration, the algorithm computes one optimal alignment which takes $O(mn)$ time. Thus steps 1-10 need $O(mn \log m)$ time. The number of different pairs of boundary lines is $O(\log^2 m)$. For every pair of boundary lines, it may be necessary to compute one optimal alignment. Thus, in the worst case, steps 11-21 need $O(mn \log^2 m)$ time. The total time is $O(mn \log^2 m)$. $\square$

In practice, if m is very large, steps 6–12 should require much less time than the upper bound, since the number of boundary lines is much less than $O(\log m)$.

The above algorithm also works for searching on the α, γ plane, only that, in step 13, when we have found that the reference alignment cannot be optimal on the α, γ coordinate plane, we need to return the centroid of the remaining search space as an approximately inverse-optimal point.

4 Further Results and Open Problems

Given two sequences S and T , a *local alignment* is obtained by finding substrings S' and T' of S and T , respectively, whose optimal global alignment score is maximum over all pairs of substrings from S and T [5]. If the lengths of S and T are n and m , the score of the local alignment of S and T can be found in $O(nm)$ time [5], for fixed weights of matches, mismatches, indels and gaps.

The inverse local alignment problem without gaps ($\gamma = 0$) asks to find an inverse-optimal or approximately inverse-optimal point on the α, β coordinate plane. This problem can be solved by slightly modifying Algorithm 2 of Section 3. Note that in this case, we search over the α, β coordinate plane. Lemmas 4, 5, 6, 7 and Theorems 4, 5 all hold for the α, β plane and equations (2), (3), (4) can be modified slightly for use on this plane. Thus we modify Algorithm 2 as follows: search on the α, β plane instead of the β, γ plane; compute the local alignment instead of global alignment in steps 4, 16; in steps 6, 18 return that inverse-optimal point instead of a line; in step 22, return the approximately inverse-optimal point, the centroid of R . The analysis of Section 3, implies that this algorithm runs in $O(mn \log^2 m)$ time.

An open problem is to extend our binary search strategy into fixed-dimensional space. For example, this could lead to an efficient algorithm for inverse local alignment with gaps, which is a search problem in the three-dimensional α, β, γ space.

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A Some Properties of Centroids

In order to solve inverse optimization problems with two parameters, we rely on a useful property of centroids of convex polygons to be proved in this section.

Lemma 8. *Let S be a convex polygon and let e be any of its edges. Let S' be a triangle with base e whose interior intersects the interior of S and such that $\text{Area}(S) = \text{Area}(S')$. Then the distance from the centroid O' of S' to e is no less than the distance from the centroid O of S to e .*

Proof. Notice that, for a triangle, when the base and area are fixed, the height and the distance from its centroid to the base are fixed. Thus, if S is a triangle, the lemma holds.

Suppose S is not a triangle. Let $e = AB$ and assume the vertices of S are $AP_1..C..P_nB$, where C is farthest from AB among all vertices, see Fig. 1(a). If $S' = \triangle ABM$, then vertex M must lie on a line L that is parallel with AB and is farther from AB than vertex C , and the centroid O' must be on a line L' that is parallel to AB . Thus, in our proof we can, without loss of generality, select M as shown in Fig. 1(a); that is, M is such that AM intersects S at G and BM intersects S at H . Denote polygon $AP_1..G$ by S_2 , $GMH..C..$ by S_4 , $H..P_nB$ as S_3 , and $AG..C..HB$ as S_1 .

Since $\text{Area}(S) = \text{Area}(S_1) + \text{Area}(S_2) + \text{Area}(S_3)$, $\text{Area}(S') = \text{Area}(S_1) + \text{Area}(S_4)$, $\text{Area}(S) = \text{Area}(S')$, then $\text{Area}(S_2) + \text{Area}(S_3) = \text{Area}(S_4)$. Obviously, the distance between the overall centroid of S_2 and S_3 and AB is less than the distance between the centroid of S_4 and AB . And in S and S' , S_1 remains unchanged, so the distance between the centroid O of S and AB is less than the distance between the centroid O' of S' and AB . $\square$

Lemma 9. *Any line passing through the centroid O of a triangle S and parallel with one edge of S partitions S into a small triangle and a trapezoid, such that the ratio of the area of the small triangle and the trapezoid $4/5$.*

Proof. Let O be the centroid of triangle ABC . Let L be the line that passes through O and is parallel with BC . Let D and E be the intersections of L with edges AB and AC , respectively (Fig. 1 (b)). By the definition of centroid of triangle, $|DE|/|BC| = 2/3$. Hence, $Area(ADE)/Area(ABC) = 4/9$; that is, $Area(ADE)/Area(BCED) = 4/5$. $\square$

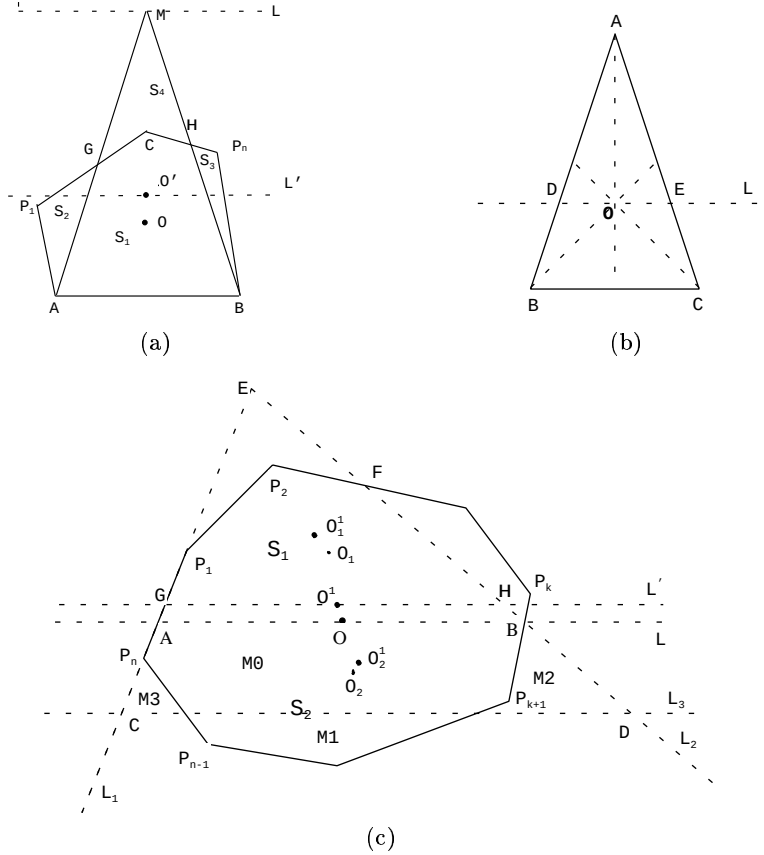


Fig. 1: Properties of the centroid of a convex polygon

Theorem 7. Let S be a convex polygon and L be a line passing through the centroid O of S . Then, the ratio of the areas of two parts of S to either side of L is between $4/5$ and $5/4$.

Proof. Let A and B be the intersection points of L with the boundary of S . Suppose the vertices of S are $P_1, \dots, P_n$ and assume, without loss of generality that A is on edge $P_n P_1$ and B is on edge $P_k P_{k+1}$. Denote by S_1 and S_2 the

regions of S to either side of L , where O_1 is the centroid of S_1 and O_2 is the centroid of S_2 . Our notation is illustrated in Fig. 1 (c). Without loss of generality, assume $Area(S_1) \leq Area(S_2)$.

First, as shown in Fig. 1 (c), reshape S_1 to a triangle $S_1^1 = EAB$ such that $Area(S_1) = Area(S_1^1)$ and reshape S_2 to a trapezoid $S_2^1 = ABDC$ which is an extension of S_1^1 and such that DC is parallel to AB and $Area(S_2) = Area(S_2^1)$. Vertex E is chosen so as to lie on the line L_1 that goes through segment AP_1 . Let L_2 be the line containing segment EB , and L_3 be the line containing CD . Denote by O_1^1 and O_2^1 the centroids of S_1^1 and S_2^1 .

Next, assume d_1 and d_1^1 are the distances from O_1 to line L and from O_1^1 to line L , d_2 and d_2^1 are the distances from O_2 to line L and from O_2^1 to line L . Now we need to show $d_1 \leq d_1^1$ and $d_2 \geq d_2^1$. If S_1 is a triangle, then no reshaping is needed, and $d_1 = d_1^1$. If S_1 is not a triangle, then, because S_1 is a convex polygon, according to Lemma 8, we know that $d_1 \leq d_1^1$. If S_2 overlaps with S_2^1 , then $d_2 = d_2^1$. Otherwise, because S_2 is a convex polygon, line L_3 partitions S_2 into two parts, $M0$ and $M1$, also S_2^1 is composed of three parts, $M1$, $M2$ and $M3$. Because the centroid of $M1$ is below the line L_3 and the centroids of $M2$ and $M3$ are above line L_3 , the centroid of S_2^1 will be nearer to line L than the centroid of S_2 . That is, $d_2 \geq d_2^1$.

Since we have $d_1 \leq d_1^1$ and $d_2 \geq d_2^1$, the centroid O^1 of the new convex polygon ECD is no lower than line L . Draw a line L' passing through O_1^1 that is parallel to L and intersects lines L_1 and L_2 at points G and H . According to Lemma 9, $Area(EGH)/Area(GHDC) = 4/5$. Because $Area(EGH) \leq Area(S_1)$ and $Area(GHDC) \geq Area(S_2)$ and $Area(S_1) \leq Area(S_2)$, we have $4/5 \geq Area(S_1)/Area(S_2) \leq 1/1$. That is, the ratio of the areas of the two parts of S to either side of L is between $4/5$ and $5/4$. $\square$